SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Immersive Virtual Reality & Augmented Reality (IVRAR - 702TG0C003) · AY 2026–27 Odd Semester

GROUP 16 TECHNICAL DOSSIER

Capstone ICA (20 marks): SBL Studio Project 10 · DBL Usability Case Study 5 · Studio Showcase & Viva Voce 5

Group / Batch	Group 16 · Batch B2 (as allotted in the finalised PBL titles sheet)
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026/tree/main/Group_16_Bhoomi_Kritivya_Tanvi_Jia
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_16_Bhoomi_Kritivya_Tanvi_Jia/docs/IVRAR_Group_16_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
How to fill this file	Use Microsoft Word (desktop recommended). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them, and do not edit this file in Google Docs: the drop-downs are how your answers are read.

Student name	Roll No.	SAP ID	Programme	Batch	Declared studio role
BHOOMI BHANDARI	I004	70122400059	B Tech (Artificial Intelligence)	B2	Triage Clinical Protocol Lead
KRITIVYA MISHRA	I037	70122400043	B Tech (Artificial Intelligence)	B2	XR Systems Architect
TANVI PAITHANKAR	I044	70122400056	B Tech (Artificial Intelligence)	B2	Spatial Telemetry & Confusion Matrix Specialist
JIA JADHAV	I069	70122400084	B Tech (Artificial Intelligence)	B2	Human Factors & Usability Engineer

Batch column = official student list. Tell Dr. Nanade by 20-Oct if any name, ID or role is wrong.

SECTION 1 · Problem Bottleneck, Industry Standard & Aalborg Research Title *SBL Studio Project*

Authorised Aalborg title	How can an AI-driven VR mass-casualty triage simulation improve START protocol categorization accuracy and reduce assessment latency for emergency medical trainees under dynamic industrial hazard conditions?
Original draft title	How can an AI-adaptive VR simulation improve triage decision-making and situational awareness for NDRF trainees responding to simulated industrial explosion scenarios?
Why the title changed	The submitted title was conceptually strong but used generic phrasing ('train to perform accurate triage under chaos'). Needed sharpening into quantitative accuracy and latency metrics.

1.1 Baseline friction & governing standard: claim verification table

Each statement below was carried over from your approved title sheet. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Standards: cite number, part and year. Statistics: cite the original study, not a blog or an AI answer.

Claim (as in title sheet)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
START (Simple Triage and Rapid Treatment) algorithm: RPM (Respiration, Perfusion, Mental Status) branches into Red, Yellow, Green, Black.	*Not yet checked: verify*	CHEMM (US HHS) START Adult Triage Algorithm, chemm.hhs.gov/startadult.htm and /startalgotext.htm (read 2026-10-25): walking, breathing, respiratory rate 30/min, perfusion by radial pulse OR capillary refill, obeys commands, tags green/yellow/red/black; original START used capillary refill, Benson, Koenig, Schultz 1996 (Prehosp Disaster Med, DOI 10.1017/S1049023X0004276X) used radial pulse. Build default = capillary refill > 2 s (2 s not on CHEMM text page; confirm), radial pulse selectable and logged separately. Training aid, not clinically validated.	Official report / documentation	Correct
Under-triage leads to preventable mortality; assessment must take < 60s/victim.	*Not yet checked: verify*	Kahn et al., Ann Emerg Med 2009;54(3):424-30, DOI 10.1016/j.annemergmed.2008.12.035 (148 pts: 3 under-, 79 over-triaged; no mortality/time data). Time: ~30 s (max 1 min) per IAVOAD START guide. Corrected.	Peer-reviewed paper (DOI)	Correct

1.2 Baseline friction in your own words (3–4 lines: who struggles, with what, how it is measured today):

Responders must triage many victims in about 30 s each, without equipment. In a 148-patient train-crash review, START under-triaged 3 and over-triaged 79 (Kahn 2009, DOI above). Our trainee baseline is NOT yet measured (Section 4).

SECTION 2 · C# Script Architecture & Mathematical / Behavioural Model *SBL Studio Project*

C# scripts in your monorepo folder today	STARTTriageSimulationManager.cs · TriageTelemetryLogger.cs · STARTEvaluator.cs · TriageSession.cs · TriageMetrics.cs · TriageCsv.cs · TriageScenario.cs · CasualtyProfile.cs · CasualtySelectable.cs · TriageUI.cs · TriageHazardEscalation.cs · TriageDesktopInput.cs · TriageTypes.cs (Assets/_Project/Scripts/Runtime/Triage/)
Telemetry logger named in the title sheet	TriageTelemetryLogger.cs
Core state machine	Casualty approached → Walk? → Breathing? → RR > 30? → Cap refill > 2 s? → Obeys commands? → Tag (Green / Yellow / Red / Black)
Fundamental equations	Triage accuracy = correct tags / casualties Under-triage rate = (true Red tagged Yellow/Green) / true Red Assessment latency per casualty (s)
Toolchain (title sheet)	Unity 2022.3 LTS URP; Industrial explosion site with 10 casualty models holding C# CasualtyProfile components; virtual triage belt with 4 color tags; C# TriageTelemetryLogger.cs.

2.1 Authorship of each script

Several scripts were provided as starter scaffolds. Be honest: mark what you wrote, what you changed, and what you used as given. Viva questions will target the code you claim. DRAFT NOTE: the casualty scenario is scripted (fixed findings, cosmetic hazard escalation); there is no machine-learning or adaptive AI component. The triage code in this repository was drafted with an AI assistant (Claude) in a coding session and reviewed by the team only in part; authorship dropdowns stay unselected until each member states what they actually wrote or changed. Built with Unity 6000.3.21f1 (URP), not the 2022.3 LTS in the title sheet; casualties are lying/standing models in an industrial yard fire with colour tags, not a belt. PENDING = PENDING — TEAM INPUT REQUIRED.

Script	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
STARTTriageSimulationManager.cs	Select ▾	Select ▾	AI-drafted (Claude). PENDING — TEAM INPUT REQUIRED
TriageTelemetryLogger.cs	Select ▾	Select ▾	AI-drafted (Claude). PENDING — TEAM INPUT REQUIRED
STARTEvaluator.cs	Select ▾	Select ▾	AI-drafted (Claude). PENDING — TEAM INPUT REQUIRED
TriageSession.cs	Select ▾	Select ▾	AI-drafted (Claude). PENDING — TEAM INPUT REQUIRED

2.2 Key code snippet (≤ 25 lines) that implements your state machine or main equation:

STARTEvaluator.cs Evaluate() lines 39-64 (blank lines omitted)

```
 39  if (v.canWalk)
 40      return new StartResult(TriageCategory.Green, TriageBranch.Walking, "Can walk.");
 42  if (!v.isBreathing)
 43  {
 44      if (!airwayRepositioned)
 45          return new StartResult(TriageCategory.Red, TriageBranch.ApneaAirwayNotAttempted, "Not
breathing: reposition the airway first.");
 46      return v.breathingAfterAirwayReposition
 47          ? new StartResult(TriageCategory.Red, TriageBranch.ApneaRespondsToAirway, "Not breathing;
breathing restored after airway repositioning.")
 48          : new StartResult(TriageCategory.Black, TriageBranch.ApneaNoResponseToAirway, "Not
breathing and no breathing after airway repositioning.");
 49  }
 51  if (v.respiratoryRate > RespiratoryRateThreshold)
 52      return new StartResult(TriageCategory.Red, TriageBranch.RespiratoryRateAbove30,
 53          "Respiratory rate " + v.respiratoryRate.ToString(CultureInfo.InvariantCulture) + "/min is
above 30.");
 55  if (PerfusionFails(v, criterion))
 56      return new StartResult(TriageCategory.Red, TriageBranch.PoorPerfusion,
 57          criterion == PerfusionCriterion.CapillaryRefill
 58              ? "Capillary refill " + v.capillaryRefillSeconds.ToString("0.#",
CultureInfo.InvariantCulture) + " s is above 2 s."
 59              : "Radial pulse absent.");
 61  if (!v.followsCommands)
 62      return new StartResult(TriageCategory.Red, TriageBranch.CannotFollowCommands, "Cannot follow
simple commands.");
 64  return new StartResult(TriageCategory.Yellow, TriageBranch.AllChecksPassed, "Breathing, perfused
and obeys commands but cannot walk.");
```

2.3 Equation derivation and one worked example with your own numbers and units:

Derivation. For N classified casualties with ground-truth tag g_i (from STAREvaluator.GroundTruth) and assigned tag a_i: accuracy = (1/N) · Σ [a_i = g_i] = correct / classified. Under-triage rate = U / R, where R = number of casualties with g_i = Red and U = those among them with a_i ∈ {Yellow, Green}. Latency of casualty i: L_i = t_tag,i − t_select,i (s), where t_select is the first time the trainee selects that casualty and t_tag the moment the tag is assigned. Mean = (1/n) Σ L_i and median over the n valid latencies. A zero denominator (N = 0 or R = 0) is written as NA, never 0.

CSV mapping. accuracy, casualties_classified, correct, under_triage_rate, true_red_total, true_red_under_triaged, mean_latency_s, median_latency_s (summary file); assessment_latency_s, ground_truth, assigned, is_correct, true_red_under_triaged (records file).

ILLUSTRATIVE worked example (invented numbers, not measured data). A session tags 5 casualties: 4 are true Red, 1 is true Yellow. The trainee tags two true Reds as Red, one as Yellow, one as Green, and the true Yellow as Red. correct = 2, so accuracy = 2/5 = 0.40. R = 4, U = 2, so under-triage rate = 2/4 = 0.50. Latencies 2, 4, 6, 8, 10 s give mean = 30/5 = 6.0 s and median = 6.0 s. The same case is checked by the unit test Metrics_AccuracyUnderTriageAndLatency_AreCorrect.

30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · GitHub Repository Verification & Git LFS Audit *SBL Studio Project*

Repository (monorepo)	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026
Your group folder	Group_16_Bhoomi_Kritivya_Tanvi_Jia/
Filled dossier location	Group_16_Bhoomi_Kritivya_Tanvi_Jia/docs/IVRAR_Group_16_Technical_Dossier.docx and .pdf
Release tag	group-16-v1.0 · build: Quest .apk or PCVR .exe

Final commit hash (full 40 characters)	f05441073ed9c3fc794deb61c0e17f1da2e487dc (code commit on feature/factory-rescue-vr, team repo bhoomibhandari88/VR-Disaster-Simulator; the instructor monorepo has NOT been updated)
Release build link (.apk / .exe)	PENDING — TEAM INPUT REQUIRED (release not published)
Root .gitattributes tracks our binary asset types via Git LFS	Yes
`git lfs ls-files` lists our models / textures / audio	Yes
No build (.apk / .exe) is committed to Git history	Yes
Release build installs and runs on a clean device	Select ▾
Filled dossier (.docx + .pdf) committed to docs/	Select ▾

3.1 Repository telemetry for your folder (run from the repo root; type the numbers)

Metric	Command	Value
Commits touching your folder	git log --oneline -- Group_16_Bhoomi_Kritivya_Tanvi_Jia/ | wc -l	21 (whole repo, all authored "Claude")
Branches used by your group	git branch -r --list '*group-16*' | wc -l	8 (2 local + 6 remote-tracking refs)
Lines of C# code	find Group_16_Bhoomi_Kritivya_Tanvi_Jia -name '*.cs' | xargs cat | wc -l	10087 (Assets/_Project)
Total folder size (MB)	du -sm Group_16_Bhoomi_Kritivya_Tanvi_Jia	663 (working tree excl. Library/Builds/.git)
LFS-tracked files in your folder	git lfs ls-files | grep Group_16_Bhoomi_Kritivya_Tanvi_Jia | wc -l	249
Raw telemetry CSV files (real participants)	ls Group_16_Bhoomi_Kritivya_Tanvi_Jia/telemetry/*.csv | wc -l	0 (no participant data exists)

3.2 Individual contribution (from git history; this table is checked against GitHub)

Member	GitHub username	Commits	% effort (agreed)	Main contribution (files / features)
Bhoomi Bhandari (I004)	PENDING	PENDING	PENDING	PENDING — TEAM INPUT REQUIRED
Kritivya Mishra (I037)	PENDING	PENDING	PENDING	PENDING — TEAM INPUT REQUIRED
Tanvi Paithankar (I044)	PENDING	PENDING	PENDING	PENDING — TEAM INPUT REQUIRED
Jia Jadhav (I069)	PENDING	PENDING	PENDING	PENDING — TEAM INPUT REQUIRED

SECTION 4 · Human-in-the-Loop Usability Evaluation & Empirical Data Pack *DBL Usability Case Study*

Primary task metric	Accuracy (%), under-triage (%), 4×4 confusion matrix
Participants	Minimum N ≥ 10 real people (your PBL dossier recommends N = 20). Voluntary, adult, written consent; anonymised IDs (P01, P02…).
Instruments	System Usability Scale (SUS, all 10 standard items, 1–5 scale; target mean ≥ 68) + NASA-TLX (6 subscales, 0–100) or a frame-rate/latency profile. Report Simulator Sickness (SSQ) if users wear a headset.
SUS scoring	Odd items: score − 1. Even items: 5 − score. Sum × 2.5 = SUS (0–100) per participant.

Real participants tested (N)	PENDING
Study design	Select ▾
Testing dates (from – to) and place	PENDING — TEAM INPUT REQUIRED
Raw CSV path in repo	PENDING — TEAM INPUT REQUIRED

Measure	N	Mean	SD	Median	Condition / comparison
SUS score	PENDING	PENDING	PENDING	PENDING	PENDING — TEAM INPUT REQUIRED
NASA-TLX (raw)	PENDING	PENDING	PENDING	PENDING	PENDING — TEAM INPUT REQUIRED
Primary task metric	PENDING	PENDING	PENDING	PENDING	PENDING — TEAM INPUT REQUIRED

[SUS box-plot (per condition)]	[Task-metric / latency plot]

DATA PROVENANCE DECLARATION (mandatory). Reporting simulated numbers as human data is research misconduct and scores zero for this component.

Every number in Section 4 comes from real people who tested our build	Select ▾
No values come from sample / scaffold / generated files (sample_experiment_data.csv, *_benchmark.csv, generate_paper_figures.py)	Select ▾
Every participant signed a consent form and we hold the forms	Select ▾
Our plots re-generate from the raw CSV in the repo	Select ▾

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Studio Showcase & Viva Voce*

LinkedIn post URL	PENDING — TEAM INPUT REQUIRED
Video length (mm:ss)	PENDING
Every member appears and speaks in the video	Select ▾

5.1 AI Interaction Audit Log: three cases where AI was wrong, made something up, or was unverifiable

Examples from this course: a fake 'Verizon 2024 XR' report, a fictional 'NEXORA Corp', a correct paper given the wrong journal. Show how you checked each one with the standard, the documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	Select ▾	PENDING — TEAM INPUT REQUIRED	Select ▾	PENDING — TEAM INPUT REQUIRED	Select ▾	Select ▾
AI-02	Select ▾	PENDING — TEAM INPUT REQUIRED	Select ▾	PENDING — TEAM INPUT REQUIRED	Select ▾	Select ▾
AI-03	Select ▾	PENDING — TEAM INPUT REQUIRED	Select ▾	PENDING — TEAM INPUT REQUIRED	Select ▾	Select ▾

5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Build	Data	Honesty	Clarity	Best moment (mm:ss)	One question for the viva
Group 17: Samarth, Ishan, Aarush, Sachi	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	PENDING — TEAM INPUT REQUIRED	PENDING — TEAM INPUT REQUIRED
Group 07: Shirin, Khushi, Pranjal	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	PENDING — TEAM INPUT REQUIRED	PENDING — TEAM INPUT REQUIRED

SECTION 6 · Individual Learning Record *not marked · one row per member · answer honestly*

These answers do not change your marks. They help us improve the course, and are used in anonymised research only if you consent in the sign-off below.

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	Before this course I had used…	Before this course I had tested with real users	I learned a skill from a teammate of another programme	Share of my writing first drafted with AI	Wrong AI facts / citations I personally caught
Bhoomi Bhandari (I004)	PENDING — TEAM INPUT REQUIRED	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾
Kritivya Mishra (I037)	PENDING — TEAM INPUT REQUIRED	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾
Tanvi Paithankar (I044)	PENDING — TEAM INPUT REQUIRED	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾
Jia Jadhav (I069)	PENDING — TEAM INPUT REQUIRED	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾

Member	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Bhoomi Bhandari (I004)	PENDING — TEAM INPUT REQUIRED	PENDING — TEAM INPUT REQUIRED
Kritivya Mishra (I037)	PENDING — TEAM INPUT REQUIRED	PENDING — TEAM INPUT REQUIRED
Tanvi Paithankar (I044)	PENDING — TEAM INPUT REQUIRED	PENDING — TEAM INPUT REQUIRED
Jia Jadhav (I069)	PENDING — TEAM INPUT REQUIRED	PENDING — TEAM INPUT REQUIRED

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, bottleneck, verified standard
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System design	Section 2: architecture, state machine, equations
IV. Evaluation	Section 4: protocol, N, SUS/TLX/metric, plots
V. Discussion & limitations	Sections 3–5: reproducibility (repo, release), AI audit lessons
Target venues (title sheet)	A: IEEE INDICON (IEEE India Council International Conference) · B: iLRN (Immersive Learning Research Network)

Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) the code in our folder runs and the Release build installs; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Bhoomi Bhandari (I004)	Select ▾	Select ▾	Select ▾		Select ▾
Kritivya Mishra (I037)	Select ▾	Select ▾	Select ▾		Select ▾
Tanvi Paithankar (I044)	Select ▾	Select ▾	Select ▾		Select ▾
Jia Jadhav (I069)	Select ▾	Select ▾	Select ▾		Select ▾

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*